RESEARCH ARTICLE

Corrosion assessment in aluminum pipe based on nonlinear ultrasonic technique using macro fiber composite transducers

Rong Wang[1,2], **Hanqi Zhang**[2,3], **Heng Chen**[1], **Yahong Wu**[1], **Ke Xiong**[2], **Qi Wu**[2]*, **Liqing Zou**[2]

1 College of Traffic Engineering, Nanjing University of Industry Technology, Nanjing, China, **2** State Key Laboratory of Mechanics and Control for Aerospace Structures, Nanjing University of Aeronautics and Astronautics, Nanjing, China, **3** School of Mechanical and Electronic Engineering, Nanjing Forestry University, Nanjing, China

* wu.qi@nuaa.edu.cn

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Abstract

Pipe corrosion, specifically pitting corrosion, is the main cause of destructive pipe leakage, driven by the harsh working environment of liquid and gas transportation. Therefore, detecting pitting corrosion is essential for ensuring the safe operation of metal pipes. This study investigates a nonlinear ultrasonic technique using macro fiber composite transducers, aiming to assess pitting corrosion in metal pipes at an early stage, with a focus on characterizing the influence of temperature on the ultrasonic nonlinearity. Macro fiber composite transducers with flexibility and high ultrasonic performance were used to actuate and detect ultrasonic guided waves propagating in pipes with curved surfaces. Considering the multi-mode propagation characteristics, the 1.4 MHz second-harmonic ultrasonic component generated by the nonlinear interaction between ultrasonic guided waves and pitting corrosion was extracted. Repeated experiments revealed that both the second-harmonic amplitude and relative nonlinear parameter exhibited a monotonic increase with the number of cycles and area of pitting corrosion. For comparison with the nonlinear results, statistical metrics including the mean slopes, coefficient of determination, and relative standard deviation of the linear fitting parameter were determined alongside the linear ultrasonic experiments. These results indicate that, despite some inherent data variability, the proposed nonlinear ultrasonic technique exhibits comparatively better sensitivity, goodness-of-fit and repeatability than linear ultrasonic methods for identifying pitting corrosion. Thus, the proposed nonlinear ultrasonic technique using macro fiber composites offers a promising complementary alternative for early corrosion assessment in metal pipes.

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Introduction

Metal pipes are critical for transporting natural gas, fuel oil, chemicals, and water because of their high specific strength and overall reliability [1, 2]. However, metal pipes are susceptible to corrosion damage, which can lead to structural failure during long-term service [3, 4]. Material corrosion can be classified into localized corrosion and general corrosion, with the occurrence of the former being significantly higher. Among the various forms of localized corrosion, pitting corrosion is one of the most destructive and severe in pipe structures [5]. Moreover, pitting corrosion promotes secondary degradation mechanisms such as intergranular corrosion, stress corrosion cracking, and corrosion fatigue. Pitting corrosion often causes leakage accidents or even explosions, not only resulting in substantial economic losses but also posing a major threat to human safety [6]. Therefore, developing effective techniques for detecting pitting corrosion is essential.

Currently, technologies such as laser scanning, microwave inspection, pulsed eddy current, electromagnetic induction thermography, and ultrasonic detection are used for damage detection in pipe structures. Laser scanning provides high resolution and rapid detection speed but incurs high equipment costs [7]. Microwave inspection cannot detect internal damage within metallic materials [8]. While pulsed eddy current and electromagnetic induction thermography technologies overcome some of these drawbacks and can detect microdamage [9, 10], they—along with laser scanning and microwave inspection—are typically deployed as offline non-destructive testing (NDT) techniques. Consequently, they are ill-suited for real-time or online continuous monitoring of corrosion damage. These offline methods often require operational downtime and are generally limited to scheduled maintenance intervals. As a result, they cannot capture the dynamic propagation of damage during the operational life of structures, thereby increasing the risk of catastrophic failure.

In practical applications, there is a high demand for effective and efficient pipe condition assessment. Compared with other detection methods, ultrasonic detection offers distinct advantages, including high efficiency, low cost, long-range capability and adaptability to pipes of various sizes and thicknesses [11, 12]. Traditional ultrasonic methods typically evaluate damage by analyzing changes in fundamental wave parameters, such as amplitude, arrival time, attenuation, and velocity dispersion. However, parameters like arrival time are not uniquely associated with damage and can be significantly influenced by environmental factors including temperature variations, stress states, and material anisotropy. While these methods detect distributed damage such as porosity or diffuse cracking through changes in attenuation or dispersion, they are generally less sensitive than nonlinear ultrasonic methods to microdamage because the ultrasonic wavelength exceeds the damage scale [13]. Consequently, nonlinear ultrasonic methods have recently been proposed to address these limitations. These techniques can detect microdamage by analyzing the generation of higher harmonic components and side-lobe frequencies in the frequency domain [14]. Unlike conventional linear ultrasonic detection, nonlinear ultrasonic methods are highly sensitive to microdamage because they evaluate material

degradation through the nonlinear effects generated by wave–damage interactions [15]. A comprehensive comparison of the advantages and drawbacks of various detection technologies (Table 1).

Early damage in pipes can be successfully detected by analyzing nonlinear ultrasonic phenomena, including higher harmonic generation, frequency mixing responses, and shifts in harmonic frequency [16, 17]. Higher harmonic generation provides high sensitivity for identifying microdamage by detecting the damage-induced material nonlinearity, thereby improving the identification of early-stage damage [18]. To mitigate the risk of false positives, potential sources of nonlinearity including instrumentation, coupling variability, and boundary conditions must be carefully controlled. Consequently, higher harmonic generation is considered the most classical nonlinear phenomenon, making it highly suitable for practical applications. For example, Hong et al. [19] reported that second- and third-order nonlinear coefficients increased with delamination damage in lined anticorrosion pipes. Ehrlich et al. [20] developed ultrasonic technology to detect nonlinearity parameters in a welded steel pipe, reporting that high levels of nonlinearity occurred in the heat-affected zone as creep damage accumulated. Hong et al. [21] detected stress corrosion cracking in stainless steel rods using nonlinear resonant ultrasound spectroscopy under specific stress and corrosion conditions, finding a general exponential relationship between the corrosion exposure time and the nonlinear parameter. Zhong et al. [22] utilized the nonlinear characteristics of ultrasonic waves to detect the intergranular corrosion of steel tubes and reported that the nonlinear coefficient generally increased with the degree of intergranular corrosion damage. Although these studies have validated the use of nonlinear ultrasound for detecting general pipeline corrosion, research specifically addressing the pitting corrosion of metal pipes remains limited. This limitation is primarily due to the minute size of pitted area, which makes accurate detection highly challenging.

Corrosion assessment is further complicated by the geometric shape and operating conditions of the pipes. Unlike large-diameter pipes, where rigid transducers suffice and guided waves can often be approximated as plate-like waves, small-diameter pipes exhibit complex cylindrical dispersion and severe curvature. This curvature renders conventional rigid transducers ineffective due to substantial acoustic energy loss. Moreover, traditional lead–zirconate–titanate (PZT) ultrasonic transducers are inherently rigid with flat surfaces, making acoustic coupling challenging on highly curved small-diameter pipe geometries. This lack of surface contact often leads to large acoustic energy loss at the interface. Additionally, the pipe always operates in a humid environment, which affects the corrosion resistance of the transducer. To mitigate these shortcomings, researchers have recently proposed macro fiber composite (MFC) transducers. MFCs are a new type of functional composite material composed of a polymer film embedded with ceramic fibers, exhibiting a strong piezoelectric effect. Compared to traditional PZT transducers, MFCs offer significantly superior flexibility, driving ability, and corrosion resistance [23], making them a highly promising alternative for the ultrasonic detection of pipe structures.

The objective of this paper is to investigate the interaction between ultrasonic guided waves and pitting corrosion in small-diameter pipes using MFC transducers. Specifically, this study establishes the relationship between ultrasonic nonlinear characteristics and the number of cycles and area of pitting corrosion. Achieving effective detection of localized corrosion damage enables better differentiation between localized and global damage effects on ultrasonic characteristics. In

Table 1. Advantages and drawbacks of the detection techniques.

Detection technology	Advantages	Drawbacks
laser scanning	high resolution	offline, high cost
microwave inspection	centimeter-level depth detection	offline, only applied to nonmetal/composites materials
pulsed eddy current	deep damage detection	offline, limited to localized damage
electromagnetic induction thermography	rapid detection	offline, high cost
linear ultrasonic detection	long-distance detection	online, misdetection of geometrical abrupt structures
nonlinear ultrasonic detection	long-distance detection	online, complicated analysis process

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